

A superpolynomial lower bound for the bit pigeonhole principle in resolution over parities

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With substantial assistance from Claude and GPT models.
More details in Section [8.3](#).

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Abstract

We present a proposed proof that the usual bit pigeonhole principle with $n + 1$ pigeons and $n = 2^\ell$ holes requires superpolynomial-size DAG-like resolution-over-parities refutations, with no regularity or proof-depth restriction. The argument converts an arbitrary short refutation into polynomially many complete affine extension blocks admitting an ordinary polynomial-calculus refutation of degree $O(\log n)$. A simultaneous coefficient substitution, weighted by a nonzero row-linear bit polynomial, would then produce an old-base consequence of degree $O(\sqrt{n}(\log n)^{3/2})$. Matching-moment extension and a cube-supported separating functional exclude that consequence. All algebraic dependencies are proved in this draft; an appendix supplies the required chessboard homology argument. We also describe the open, versioned research framework in which the proof was developed. This is a working manuscript prepared for review, not an independently verified or formally certified result.

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1 The claim and its setting

Resolution over parities augments ordinary resolution with affine equations over $\mathbb{F}_2$. The system was introduced by Itsykson and Sokolov [IS14, IS20]. Its unrestricted proof-size lower-bound problem remains an explicit open benchmark in the literature reviewed for this draft [IPS26]. The contribution claimed here is a superpolynomial lower bound for the standard bit pigeonhole principle in unrestricted DAG-like $\text{Res}(\oplus)$.

1.1 The bit pigeonhole principle

Fix $\ell \geq 2$, put $n = 2^\ell$ and $m = n + 1$, and introduce

$$b_{it} \quad (i \in [m], t \in [\ell]).$$

The row b_i encodes one of the n labels in $\{0, 1\}^\ell$. For a label z , let $[b_{it} \neq z_t]$ denote b_{it} when $z_t = 0$ and $\neg b_{it}$ when $z_t = 1$. The usual CNF encoding is

$$\text{BPHP}_n^{n+1} = \bigwedge_{\substack{i < i' \in [m] \\ z \in \{0,1\}^\ell}} \left(\bigvee_{t \in [\ell]} [b_{it} \neq z_t] \vee \bigvee_{t \in [\ell]} [b_{i't} \neq z_t] \right). \quad (1.1)$$

It has $v = m\ell$ variables and $n \binom{m}{2}$ clauses of width 2ℓ . No separate range or pigeon axiom is required: every Boolean row encodes exactly one available label.

1.2 The proof system

A *linear clause* is a disjunction of affine equations over $\mathbb{F}_2$. The basic rules are complementary-parity resolution and semantic weakening:

$$\frac{A \vee (u = 0) \quad B \vee (u = 1)}{A \vee B}, \quad \frac{C}{D} \quad \text{if } C \models D. \quad (1.2)$$

The pivot u may be any affine form. A refutation is a finite acyclic derivation from initial clauses to the empty clause. Its size is the number of nodes, including initial nodes, in its proof DAG. Earlier nodes may be used repeatedly, and no restriction is placed on proof depth or on repeated use of related pivots.

We also consider the convention permitting every sound inference from at most two linear-clause premises. Lemma 6.4 reduces this convention to (1.2) with constant node overhead. Fresh unconstrained propositional variables, if used, may be set to constants throughout a proof. Extension axioms defining new predicates are not part of the system considered here.

Theorem 1.1 (Main theorem, proposed proof). *For every fixed real $K > 0$, there is $\ell_0(K)$ such that for every $\ell \geq \ell_0(K)$, every DAG-like $\text{Res}(\oplus)$ refutation of $\text{BPHP}_{2^\ell}^{2^\ell+1}$ has more than $2^{K\ell}$ nodes. The conclusion holds for both rule conventions above.*

Thus the minimum line count is $n^{\omega(1)}$ along powers of two. This also implies a superpolynomial lower bound on total proof-description size. The input length in (1.1) is polynomial in n , so using input length instead of n gives the same qualitative conclusion.

1.3 Prior results and the scope of the claim

Efremenko, Garlík, and Itsykson [EGI25] prove a $2^{\Omega(n^{1/3}/\log n)}$ size bound for regular $\text{Res}(\oplus)$ refutations of bit PHP. Their unrestricted DAG-like result is an affine-width bound. Bhattacharya, Chattopadhyay, and Dvořák [BCD24] exhibit formulas that are exponentially hard for bottom-regular $\text{Res}(\oplus)$, although they have short proofs in general ordinary resolution. This separation demonstrates the importance of removing regularity.

Bhattacharya and Chattopadhyay [BC25], Byramji and Impagliazzo [BI25], and the merged STOC 2026 paper [BCBI26] obtain exponential bounds in regimes allowing nearly quadratic proof depth. Their bounds retain a proof-depth restriction. Itsykson, Podolskii, and Shekhovtsov [IPS26] explicitly distinguish this progress from the general superpolynomial size problem; their unrestricted size consequence is quadratic.

The statement proposed here removes both regularity and proof-depth restrictions. The literature review found no preceding theorem with that scope, but it is not an exhaustive certification of novelty. The full $\text{AC}^0[p]$ -Frege lower-bound problem remains outside the conclusion of Theorem 1.1.

1.4 Proof organization

The proof has three interfaces. First, matching moments give a stable ordinary-degree filtration for functional unary PHP, and a separate signed-direction construction detects every nonzero row-linear bit polynomial. Second, a restriction-dimension argument supplies a common polynomial for all high-rank affine blocks, while low-rank blocks admit exact coefficient packing. A single weighted substitution removes the whole affine family. Third, each source proof clause is assigned a complete extension block; local PC derivations simulate the rules without a proof-height cost.

Sections 2–6 prove these interfaces in full. Section 7 closes the parameters. Appendix A includes the precise homological input behind the matching extension. A companion online research notebook records the development of the argument and supporting experiments [Repo]; Section 8 describes these supplementary materials. The proofs presented here can be read independently of that notebook: none of its lemmas is required as an unexpanded black box in this draft.

2 Ordinary degree and polynomial calculus

All polynomial rings in the main argument are over $\mathbb{F}_2$. Degree bounds and accuracy parameters are integers. Polynomials are ordinary commutative polynomials. In particular, powers are not silently reduced using Booleanity. The Boolean equations are

$$\text{BOOL}_x = \{x_j^2 - x_j : j\}.$$

Every displayed generator is an equation with right side zero.

For a finite polynomial system $\mathcal{F}$, write

$$\mathcal{I}_B(\mathcal{F}) = \text{span}_{\mathbb{F}_2}\{qF : F \in \mathcal{F}, \deg(qF) \leq B\} \subseteq \mathcal{P}_{\leq B}. \quad (2.1)$$

The degree constraint is imposed on each generator multiple. Cancellation between larger-degree multiples does not make them legal in $\mathcal{I}_B$.

Ordinary PC derives axioms of $\mathcal{F}$, linear combinations of previous lines, and $x_j g$ from a previous line g . A degree- B derivation has no line of degree exceeding B . Let $\mathcal{C}_B(\mathcal{F})$ be the space of polynomials derivable in this way. Refutation means deriving one. The space $\mathcal{C}_B$ is closed under linear combinations, but equality with $\mathcal{I}_B$ requires a proof.

Lemma 2.1 (Multiplying a completed line). *If g has a degree- B PC derivation and q is any polynomial, then qg has a PC derivation through $\max\{B, \deg q + \deg g\}$. Moreover $\mathcal{I}_B(\mathcal{F}) \subseteq \mathcal{C}_B(\mathcal{F})$.*

Proof. Expand q into monomials. Starting from the already derived line g , multiply by the variables of each monomial one at a time, and then take the indicated linear combination. Each new line has degree at most $\deg g + \deg q$. The zero case is immediate. Applying this construction to each legal axiom multiple in (2.1), then summing, proves the second assertion. There is no need to multiply the earlier derivation of g . $\square$

Lemma 2.2 (Degree-controlled Boolean reduction). *For every polynomial R of degree at most d , its squarefree remainder $\bar{R}$ satisfies*

$$R - \bar{R} \in \mathcal{I}_d(\text{BOOL}_x).$$

If R vanishes at every Boolean point, then $R \in \mathcal{I}_d(\text{BOOL}_x)$. In particular, if $\deg q \leq k$, then

$$q^2 - q \in \mathcal{I}_{2k}(\text{BOOL}_x).$$

Proof. For $a \geq 2$,

$$x^a - x = (x^2 - x)(1 + x + \cdots + x^{a-2}).$$

Apply this identity to the powers in each monomial of R , one variable at a time. Every generator multiple has degree at most the original monomial degree, so the first assertion follows.

A squarefree polynomial that vanishes on $\{0, 1\}^v$ is zero. For example, write it as $A + x_v B$ with A, B independent of x_v . Evaluations at $x_v = 0, 1$, followed by induction on v , give $A = B = 0$. This proves the second assertion. Finally, every value of q at a Boolean point belongs to $\mathbb{F}_2$, where its square equals itself; apply the second assertion at degree $2k$. $\square$

We use this reduction lemma only when the required degree is explicitly available. The high-rank step below instead needs literal membership in an affine ideal; pointwise vanishing on a finite field does not by itself supply that stronger conclusion.

3 Matching moments and a stable old filtration

For $m, N \geq 1$, the functional unary system $\mathcal{F}_{m,N}^{\text{fun}}$ has variables X_{ij} and generators

$$\begin{aligned} & X_{ij}^2 - X_{ij}, \\ & X_{ij}X_{ij'} & (j \neq j'), \\ & X_{ij}X_{i'j} & (i \neq i'), \\ & \rho_i - 1, \quad \rho_i = \sum_{j=1}^N X_{ij}. \end{aligned}$$

The row exclusions in the second line are part of this auxiliary system. They are not inferred from a weaker unary encoding.

A matching T is a set of cells with distinct rows and distinct columns; write $X_T = \prod_{(i,j) \in T} X_{ij}$. The empty matching corresponds to the constant one.

Lemma 3.1 (Complete moment equations). *A linear functional on $\mathcal{P}_{\leq B}$ annihilating $\mathcal{I}_B(\mathcal{F}_{m,N}^{\text{fun}})$ is exactly a choice of matching moments z_T , $|T| \leq B$, satisfying*

$$\sum_{j \notin \text{col}(T)} z_{T \cup \{(i,j)\}} = z_T \quad \text{for } |T| < B, i \notin \text{row}(T). \quad (3.1)$$

Its value on an ordinary monomial is the moment of its squarefree support if that support is a matching, and is zero otherwise. Normalization is the additional condition $z_{\emptyset} = 1$.

Proof. Boolean equations reduce powers within the original degree. A squarefree monomial whose support has a repeated row or column contains one of the exclusion generators as a factor. Thus every polynomial reduces, within its degree, to a linear combination of matching monomials. Conversely, the stated rule for evaluating ordinary monomials annihilates every legal Boolean or exclusion multiple.

It remains to check row equations. In a row-equation multiple $q(\rho_i - 1)$ of degree at most B , we may first put q into the matching normal form: any error is a Boolean or exclusion consequence through $B - 1$, and multiplication by $\rho_i - 1$ keeps its certificate within B . For a matching multiplier X_T , if row i is already occupied, the product reduces to zero. Otherwise its reduction is

$$\sum_{j \notin \text{col}(T)} X_{T \cup \{(i,j)\}} - X_T.$$

These are precisely (3.1), and they exhaust the constraints. No positivity or multiplicativity is imposed on the functional. $\square$

Let $\Delta_{s,N}$ be the chessboard complex whose faces are matchings on an s -row, N -column board. The following consequence of the chessboard connectivity theorem of Björner, Lovász, Vrećica, and Živaljević [BLVZ94] is proved, with its homological prerequisite, in Appendix A:

$$N \geq 2s - 1 \implies \tilde{H}_{s-2}(\Delta_{s,N}; \mathbb{F}_2) = 0. \quad (3.2)$$

Theorem 3.2 (Extension of prescribed matching moments). *Suppose $m \geq B$, $N \geq 2B - 1$, and $0 \leq k \leq B$. Every functional annihilating $\mathcal{I}_k(\mathcal{F}_{m,N}^{\text{fun}})$ extends to one annihilating $\mathcal{I}_B(\mathcal{F}_{m,N}^{\text{fun}})$. The constant moment is preserved. In particular normalized functionals exist through degree B .*

Proof. Use Lemma 3.1. Starting from the prescribed moments, fill degrees $s = k + 1, \dots, B$. At degree one choose moments in each row whose sum is the prescribed constant moment; a column exists because $N \geq 1$.

For $s \geq 2$, fix a set S of s rows. Form the reduced $(s - 2)$ -chain

$$z_S = \sum_{\substack{T \text{ a matching on } S \times [N] \\ |T| = s-1}} z_T [T]$$

in $\Delta_{s,N}$. This is a cycle. At $s = 2$, its augmentation is the sum of two equal row totals, which is zero in $\mathbb{F}_2$. For $s > 2$, a face U of size $s - 2$ misses two rows of S . Its boundary coefficient is the sum of the prescribed marginal sums for those two rows, hence $z_U + z_U = 0$.

Since $N \geq 2s - 1$, equation (3.2) gives an $(s - 1)$ -chain y_S with boundary z_S . Assign the coefficients of y_S as the new moments on full s -matchings of $S \times [N]$. For a fixed $(s - 1)$ -matching, its boundary equation is exactly the missing-row equation in (3.1).

Different s -row sets have disjoint new unknowns: a size- s matching has a unique occupied row set. Their shared lower faces retain their already assigned values. Thus these fillings are globally consistent and complete the induction. The argument works for either value of the constant moment. Starting with $z_\emptyset = 1$ at degree zero supplies a normalized functional. $\square$

Corollary 3.3 (Old filtration stability and PC closure). *Under $m \geq B$ and $N \geq 2B - 1$,*

$$\mathcal{I}_B(\mathcal{F}_{m,N}^{\text{fun}}) \cap \mathcal{P}_{\leq k} = \mathcal{I}_k(\mathcal{F}_{m,N}^{\text{fun}}) \quad (0 \leq k \leq B), \quad \mathcal{C}_B(\mathcal{F}_{m,N}^{\text{fun}}) = \mathcal{I}_B(\mathcal{F}_{m,N}^{\text{fun}}). \quad (3.3)$$

In particular $1 \notin \mathcal{C}_B(\mathcal{F}_{m,N}^{\text{fun}})$.

Proof. If a polynomial of degree at most k is outside $\mathcal{I}_k$, finite-dimensional linear duality gives a functional vanishing on $\mathcal{I}_k$ but not on that polynomial. Extend it by Theorem 3.2. The polynomial is then outside $\mathcal{I}_B$ as well. The reverse inclusion in the first equality is immediate.

To prove PC closure, induct over a degree- B derivation. Axioms and linear combinations lie in $\mathcal{I}_B$. At a nonzero multiplication step $g \mapsto X_{ij}g$, the predecessor has ordinary degree at most $B - 1$. If $g \in \mathcal{I}_B$, the first equality puts it in $\mathcal{I}_{B-1}$. Multiplication of its legal generator multiples by X_{ij} stays in $\mathcal{I}_B$. Hence $\mathcal{C}_B \subseteq \mathcal{I}_B$; the other direction is Lemma 2.1. A normalized annihilator excludes one. $\square$

The equalities in (3.3) concern the old functional system. We will not use an analogous assertion for a system augmented with extension blocks.

4 The bit decoder and a separating functional

Continue with $n = 2^\ell$, $\ell \geq 2$, and $m = n + 1$. The old compact bit system is

$$\mathcal{Q}_n = \{E_{ii'} : i < i' \in [m]\} \cup \text{BOOL}_b, \quad E_{ii'} = \prod_{t=1}^{\ell} (1 - b_{it} - b_{i't}). \quad (4.1)$$

The degree of each collision generator is ℓ . Its vanishing asserts that the two bit rows encode different labels.

Identify unary columns with labels $z \in \{0, 1\}^\ell$, and define the ordinary ring map

$$\tau(b_{it}) = \sum_{z: z_t=1} X_{iz}. \quad (4.2)$$

Lemma 4.1 (Two-row interpolation). *Let $\mathcal{U}_{i,i'}$ consist of Booleanity, same-row exclusions, and the two row-sum equations for distinct rows i, i' . For a polynomial R supported on these rows and of degree at most d ,*

$$R - \sum_{z,w} R(e_z, e_w) X_{iz} X_{i'w} \in \mathcal{I}_{\max\{d, 2\}}(\mathcal{U}_{i,i'}).$$

Here e_z, e_w are the corresponding one-hot assignments. Column exclusions are not used in this interpolation.

Proof. Reduce powers and same-row collisions. The remainder is a linear combination of 1, individual cells of the two rows, and products containing one cell from each row. This reduction costs at most d . Replace the constant by $\rho_i \rho_{i'}$, using

$$1 - \rho_i \rho_{i'} = (1 - \rho_i) + \rho_i (1 - \rho_{i'}),$$

and replace each remaining single cell by its product with the other row sum. These changes have certificates of degree at most two. The result is bilinear, and its coefficient at $X_{iz}X_{i'w}$ is its value at (e_z, e_w) , namely $R(e_z, e_w)$. $\square$

Lemma 4.2 (Degree-preserving old-base transfer). *If $q \in \mathcal{C}_B(\mathcal{Q}_n)$, then*

$$\tau q \in \mathcal{C}_B(\mathcal{F}_{m,n}^{\text{fun}}).$$

The map τ is injective and preserves ordinary degree.

Proof. The bit Booleanity images have explicit degree-two certificates:

$$\tau(b_{it}^2 - b_{it}) = \sum_{z: z_t=1} (X_{iz}^2 - X_{iz}).$$

For a collision image, use Lemma 4.1 with $R = \tau(E_{ii'})$. At (e_z, e_w) its value is one exactly when $z = w$. Since $\ell \geq 2$, the lemma gives

$$\tau(E_{ii'}) - \sum_z X_{iz}X_{i'z} \in \mathcal{I}_\ell(\mathcal{U}_{i,i'}).$$

The remaining sum consists of degree-two column exclusions. Thus every old bit axiom image has an NS, hence PC, certificate through its original degree.

For each row, the coefficient vectors of the ℓ forms $\tau(b_{it})$ are linearly independent: the column labeled by the t -th unit bit vector isolates the t -th coefficient. Complete these forms to a linear coordinate system in the unary variables. This proves injectivity and degree preservation. Finally, replay a PC derivation under τ . A variable multiplier becomes a linear polynomial, which is implemented by variable multiplications and a linear combination at the same degree. The certified axiom images initialize this replay. $\square$

Let $\mathcal{L}_k$ be the space spanned by monomials of degree at most k containing at most one bit variable from each row. Its dimension is

$$\dim \mathcal{L}_k = \sum_{j=0}^k \binom{m}{j} \ell^j \quad \text{when } k \leq m. \quad (4.3)$$

Theorem 4.3 (Row-linear separation through a larger PC degree). *Suppose $1 \leq k \leq B$, $4(k-1) < n$, and $n \geq 2B - 1$. For every nonzero $f \in \mathcal{L}_k$,*

$$\tau f \notin \mathcal{C}_B(\mathcal{F}_{m,n}^{\text{fun}}), \quad f \notin \mathcal{C}_B(\mathcal{Q}_n).$$

Proof. If f is constant, it is one, and a normalized functional from Theorem 3.2 separates it. Otherwise choose a degree- t monomial with nonzero coefficient, where $t = \deg f \leq k$. Let S be its row set and let a_i be its bit coordinate in row $i \in S$.

Choose disjoint two-point coordinate lines

$$C_i = \{c_i, c_i + e_{a_i}\} \subseteq \{0, 1\}^\ell, \quad i \in S.$$

For any required direction there are $n/2$ possible lines. The points in previously chosen lines forbid at most $2(t-1)$ of them. The inequality $4(k-1) < n$ therefore permits greedy selection. Let $F = \bigcup_{i \in S} C_i$, so $|F| = 2t$.

On the $m - t$ remaining rows and the $n - 2t$ labels outside F , choose a normalized matching functional λ through degree $B - t$. Theorem 3.2 applies because

$$m - t \geq B - t, \quad n - 2t \geq 2(B - t) - 1.$$

There are residual labels: since n is divisible by four, the strict packing inequality implies $k \leq n/4$. At degree zero, λ is simply the normalized functional on constants.

Define moments $\mu(X_T)$ for all matchings of size at most B . Give a matching value zero unless it occupies every row $i \in S$ at a label in C_i , and every other occupied row uses a label outside all of F . In the remaining case set

$$\mu(X_T) = \lambda(X_{T|_{[m] \setminus S}}). \quad (4.4)$$

Extend this to ordinary monomials by Boolean power reduction and value zero on row or column collisions.

We verify all equations of Lemma 3.1. An already incorrect selected-row label or an outside row using a label in F cannot be repaired by adding a cell. If a matching misses at least two selected rows, one extension leaves its value zero. If it misses exactly one selected row and the marginal adds that row, its two allowed endpoints give equal residual values whose sum is zero in $\mathbb{F}_2$. A marginal adding a different row still has value zero. Finally, if every selected row is present, a marginal in another row is a marginal of λ , since all added labels in F have value zero. These cases exhaust the equations. Consequently μ annihilates $\mathcal{I}_B$, and hence $\mathcal{C}_B$ by Corollary 3.3.

Only monomials of f with row set S can contribute to $\mu(\tau f)$. They have degree t and use one bit in each such row. Equation (4.4) evaluates them by summing over $\prod_{i \in S} C_i$, with residual value $\lambda(1) = 1$. In a selected row the sum of bit a_i over its two endpoints is one, whereas the sum of every other bit is zero. The product of these one-bit differences isolates exactly the chosen monomial's coefficient. Thus $\mu(\tau f) = 1$, proving the first exclusion. The second follows from Lemma 4.2. $\square$

The functional μ in the nonconstant case has value zero on one. It is a separating direction in a finite-dimensional dual space, not a probability distribution. Its role is to exclude the particular nonzero f , rather than merely to exclude a refutation of the old base.

5 Excluding one complete affine extension level

The product construction below belongs to the extension-Nullstellensatz approach developed by Buss, Impagliazzo, Krajíček, Pudlák, Razborov, and Sgall [BIKPRS]. We use a complete one-level family and give the needed algebraic arguments explicitly.

An accuracy- h affine block has a finite tuple of old affine polynomials $g_1, \dots, g_s$, fresh coefficient variables r_{ui} , and product

$$P = \prod_{u=1}^h \left(1 - \sum_{i=1}^s r_{ui} g_i \right). \quad (5.1)$$

Its *complete* axiom set contains every $g_i P$ and every $r_{ui}^2 - r_{ui}$. Products $P = 0$ are not added as axioms. Different blocks have disjoint coefficient variables, and every input is affine in the old bits alone.

5.1 Cleanup and ordinary affine restriction

A block is *proper* if its affine input span is nonzero and does not contain the constant one. Its rank is the dimension of that span.

Lemma 5.1 (Scalar cleanup). *Zero-span and unit-span blocks can be removed by substituting constants for their coefficients, without increasing PC degree or changing the other blocks. After zero inputs are discarded, every companion in a proper block has ordinary degree $2h + 1$.*

Proof. For zero span all inputs are zero; set its coefficients to zero. For unit span choose constants c_i with $\sum_i c_i g_i = 1$. Set the first coefficient row to c_i and all other rows to zero, making $P = 0$. Every axiom of an eliminated block maps to zero, and coefficient domains map to valid constants. Other blocks use only old bits and are unchanged. A constant substitution preserves a PC derivation and its degree.

In a proper span each nonzero input has nonzero linear part; otherwise the span contains one. Each factor of (5.1) then has nonzero degree-two part, because its own coefficient variables are distinct. Their product has degree $2h$ in the ordinary polynomial ring. Multiplying by a nonzero affine input gives degree $2h + 1$. Coefficients for an identically zero input may independently be set to zero, so those inputs cause no exceptional axiom degree. $\square$

For a proper block, an affine basis $F_1, \dots, F_r$ of its input span has independent linear parts. Indeed, a nontrivial relation among those parts would give either a dependence among the F_j or a nonzero constant in their span. Thus its zero set is a nonempty affine flat of codimension r .

Lemma 5.2 (A common ordinary restriction kernel). *Let $h = 3\ell$, $m = n + 1$, $v = m\ell$, and let $M \geq 1$ bound the number of proper blocks. Put*

$$k = \left\lceil \sqrt{m \ln(4M)} \right\rceil$$

and suppose $k \leq m$. For all blocks of rank greater than $h(k + 1)$, there is a common nonzero $f \in \mathcal{L}_k$ whose restriction to each zero flat is the zero ordinary polynomial in affine free coordinates. For each such block it has a literal representation

$$f = \sum_i a_i g_i, \quad \deg a_i \leq k - 1. \quad (5.2)$$

Proof. Restriction to a codimension- r affine flat maps $\mathcal{L}_k$ into ordinary polynomials of degree at most k on $v - r$ free coordinates. Its image dimension is at most $\binom{v-r+k}{k}$. Every high-rank block has $r \geq r_* = 3\ell(k + 1) + 1$. If $r_* > v$, there are no such blocks and any nonzero f suffices. Otherwise

$$\begin{aligned} \frac{\binom{v-r_*+k}{k}}{\binom{m}{k} \ell^k} &= \prod_{j=0}^{k-1} \left(1 - \frac{r_* - 1 - (\ell + 1)j}{v - \ell j} \right) \\ &\leq \left(1 - \frac{k}{m} \right)^k \leq e^{-k^2/m}. \end{aligned} \quad (5.3)$$

For the equality, order the numerator factors increasingly, as $v - r_* + 1 + j$, and the denominator factors as $v - \ell j$. The numerator of each subtracted fraction is at least ℓk ; its denominator is at most $m\ell$. The presence of a high block implies $k < m$, so all factors used here are nonnegative.

The sum of all high restriction-image dimensions is consequently at most

$$M e^{-k^2/m} \binom{m}{k} \ell^k \leq \frac{1}{4} \binom{m}{k} \ell^k < \dim \mathcal{L}_k.$$

The joint restriction map has a nonzero kernel.

For a fixed high block, complete $F_1, \dots, F_r$ to affine coordinates. The ordinary polynomial for f has zero specialization when the first r coordinates are zero. Every monomial therefore has one of these coordinates as a factor. Assign each such monomial to one of its factors, obtaining $f = \sum_j F_j A_j$ with $\deg A_j \leq k - 1$. Pull back the affine coordinate change and express each F_j as a constant linear combination of the original g_i . This gives (5.2) at the same degree. $\square$

5.2 One simultaneous weighted substitution

Lemma 5.3 (Weighted removal of the affine family). *Let an old system $\mathcal{F}$ contain all old Boolean equations, and adjoin a complete proper one-level affine family of accuracy h . Suppose it has an ordinary-PC refutation of degree $D \geq 2h + 1$. Let $k \geq 1$, and let f be a multilinear polynomial of degree at most k having representations (5.2) for every block of rank greater than $h(k + 1)$. Then*

$$f \in \mathcal{C}_{k(D+1)}(\mathcal{F}).$$

Proof. We give one substitution σ fixing the old variables and sending every coefficient variable to an old polynomial of degree at most k .

Low rank. For a block of rank $r \leq h(k + 1)$, choose an affine basis $F_1, \dots, F_r$ and partition it into h ordered groups of size at most $k + 1$, allowing empty groups. In a group J , telescoping gives

$$1 - \prod_{j \in J} (1 - F_j) = \sum_{j \in J} F_j \prod_{\substack{t \in J \\ t < j}} (1 - F_t).$$

Express the F_j in the original g_i . The coefficient of each g_i on the right has degree at most k ; assign it to the corresponding row of (5.1). The block product becomes

$$Z = \prod_{j=1}^r (1 - F_j).$$

At every Boolean point, Z is the indicator that all inputs vanish. Thus $g_i Z$ is a zero Boolean function of ordinary degree at most $r + 1$. Lemma 2.2 gives its certificate through $r + 1$. After multiplication by f , each companion image is certified through

$$r + 1 + k \leq h(k + 1) + 1 + k \leq k(2h + 2). \quad (5.4)$$

The last inequality is equivalent to $(k - 1)(h + 1) \geq 0$.

High rank. Use (5.2) as the first coefficient row and set the other rows to zero. The product becomes $1 - f$. The weighted companion image is

$$f g_i (1 - f) = -g_i (f^2 - f).$$

By Lemma 2.2, it has an old Boolean certificate through $2k + 1 \leq k(2h + 2)$.

All remaining axioms. If a coefficient variable maps to β , its weighted field equation is $f(\beta^2 - \beta)$, certified through $3k$. An old degree- e axiom F has weighted image fF , with certificate through $e + k \leq k(e + 1)$. Together with (5.4), this yields the complete ledger

Original axiom	Original degree e	Weighted image ceiling
High companion	$2h + 1$	$2k + 1$
Low companion	$2h + 1$	$r + 1 + k$
Coefficient Booleanity	2	$3k$
Old axiom	e	$e + k$

Every entry fits $k(e + 1)$. Proper-block companion degrees are the ordinary degrees justified in Lemma 5.1. All assignments use only old variables, so they define a single substitution even when input tuples overlap.

Primitive PC replay. Let $B = k(D + 1)$. Replace each source line q by $f\sigma(q)$. The preceding certificates initialize every source axiom used by the degree- D refutation through at most B . Linear combinations commute with the weighted map. For a nonzero source multiplication $q \mapsto yq$, write $d = \deg q$. Then $d + 1 \leq D$, while

$$\deg(f\sigma(q)) \leq k(d + 1), \quad \deg \sigma(y) \leq k.$$

Multiplying the completed weighted line by $\sigma(y)$, using Lemma 2.1, therefore stays within $k(d + 2) \leq B$. Zero source lines cause no difficulty. The final weighted image of one is f , proving the assertion. $\square$

Theorem 5.4 (Finite-parameter affine exclusion). *Let $n = 2^\ell$, $\ell \geq 2$, $m = n + 1$, and $h = 3\ell$. Adjoin any finite complete one-level old-affine family to $\mathcal{Q}_n$. Let $M \geq 1$ bound its number of proper blocks after scalar cleanup, and put*

$$k = \left\lceil \sqrt{m \ln(4M)} \right\rceil, \quad B = k(D + 1).$$

If

$$D \geq 2h + 1, \quad k \leq m, \quad n \geq 2B - 1, \quad 4(k - 1) < n, \quad (5.5)$$

then the augmented system has no ordinary-PC refutation through degree D . In particular this excludes polynomially many blocks at every polylogarithmic degree for sufficiently large n .

Proof. Apply scalar cleanup, which preserves a hypothetical refutation and its degree. Lemma 5.2 supplies a nonzero $f \in \mathcal{L}_k$ with the representations required by Lemma 5.3. The latter gives $f \in \mathcal{C}_B(\mathcal{Q}_n)$. Since $k \leq B$, equation (5.5) contradicts Theorem 4.3. If no proper blocks remain, the same argument may use $f = 1$, or directly use the normalized old functional.

For polynomial M , $k = O(\sqrt{n \log n})$. For any fixed polylogarithmic bound on D , $B = \sqrt{n}$ polylog $n = o(n)$, so the remaining conditions hold eventually. If the proposed degree bound is smaller than $2h + 1$, enlarge it to $2h + 1$; this remains polylogarithmic and still excludes the original refutation. $\square$

6 A direct PC simulation of the proof DAG

Represent a linear clause by its affine true-indicators:

$$C = \bigvee_i (g_{C,i} = 1), \quad Z(C) = \{b : g_{C,i}(b) = 0 \text{ for all } i\}.$$

An equation $a(b) = \alpha$ has true-indicator $1 + a(b) + \alpha$. The set $Z(C)$ is an affine subspace or empty.

Assign to each nonempty clause a fresh complete accuracy- h block, and write its product as P_C . Set $P_\emptyset = 1$, using no block for the empty clause. Fix the whole block family before following the proof DAG. Clause values $P_C = 0$ will be derived; they are not included among its axioms.

Lemma 6.1 (Prefix identity). *For every clause block there are polynomials $U_{C,i}$ satisfying*

$$1 - P_C = \sum_i U_{C,i} g_{C,i}, \quad \deg P_C \leq 2h, \quad \deg U_{C,i} \leq 2h - 1. \quad (6.1)$$

Its companions have degree at most $2h + 1$.

Proof. Put $L_u = 1 - \sum_i r_{ui} g_{C,i}$. Telescoping the product gives

$$1 - \prod_{u=1}^h L_u = \sum_{u=1}^h (1 - L_u) \prod_{w < u} L_w.$$

Thus one may take

$$U_{C,i} = \sum_{u=1}^h r_{ui} \prod_{w < u} L_w.$$

Each L_w has degree at most two, giving all the stated bounds. $\square$

Lemma 6.2 (Complementary-parity resolution). *Suppose $A = (\bigvee_i (a_i = 1)) \vee (u = 1)$ and $B = (\bigvee_j (b_j = 1)) \vee (1 - u = 1)$, with u affine. Let T have the union of the context inputs a_i, b_j . If P_A, P_B have PC derivations through degree K , then P_T has a derivation in the same fixed block system through $\max\{K, 4h + 1\}$.*

Proof. The context companions $a_i P_T, b_j P_T$ are axioms of the conclusion block; a repeated input may use the same axiom. The prefix identities yield

$$R_A = P_T - U_{A,u} u P_T = P_A P_T + \sum_i U_{A,i} (a_i P_T), \quad (6.2)$$

$$R_B = P_T - U_{B,1-u} (1 - u) P_T = P_B P_T + \sum_j U_{B,j} (b_j P_T). \quad (6.3)$$

Lemma 2.1 derives these through $\max\{K, 4h\}$. In particular it multiplies the completed lines P_A, P_B , not their entire earlier derivations.

For an affine $u = c + \sum_t c_t b_t$, the identity

$$u^2 - u = \sum_t c_t (b_t^2 - b_t)$$

has a degree-two proof. Therefore

$$(1 - u) R_A - U_{A,u} P_T (u^2 - u) = (1 - u) P_T \quad (6.4)$$

is derivable through $\max\{K, 4h + 1\}$. The right side is a completed line of degree at most $2h + 1$. Multiply it by $U_{B,1-u}$, costing at most $4h$, and add (6.3). The result is P_T . All multiplications expand into primitive PC steps by Lemma 2.1. $\square$

Lemma 6.3 (Semantic weakening and tautologies). *If $C \models D$ and P_C has a degree- K derivation, then P_D is derivable through $\max\{K, 4h\}$. A tautological D has a direct value derivation through $2h + 1$.*

Proof. If $Z(D)$ is nonempty, then $Z(D) \subseteq Z(C)$. Each $g_{C,i}$ vanishes on $Z(D)$, so affine linear algebra gives constants λ_{ij} with

$$g_{C,i} = \sum_j \lambda_{ij} g_{D,j}.$$

To see this directly, choose affine coordinates beginning with a basis of the equations defining $Z(D)$. An affine polynomial vanishing when these coordinates are zero has only those coordinate terms. Consequently $g_{C,i}P_D$ is a constant linear combination of conclusion companions. Now use

$$P_D = P_C P_D + \sum_i U_{C,i}(g_{C,i}P_D)$$

and Lemma 2.1, with ceiling $\max\{K, 4h\}$.

If $Z(D)$ is empty, its affine equations are inconsistent, and their span contains one. The same constant combination of their companions gives P_D , through $2h + 1$. If the conclusion is empty, its value is one; the preceding formulas still apply with no conclusion block. In the nonempty-zero-set case, an implication to the empty clause forces all premise indicators to be identically zero, so the already derived premise value is one. $\square$

Lemma 6.4 (Two-premise semantic inference). *Every sound inference $A \wedge B \models D$ between linear clauses over $\mathbb{F}_2$ reduces to a tautology case, one semantic weakening, or two semantic weakenings followed by one complementary-parity resolution. The last case introduces at most two auxiliary clauses.*

Proof. Put $W = Z(D)$. Soundness says $W \subseteq Z(A) \cup Z(B)$. An empty W gives a tautology, and containment in either one of the two sets gives a one-premise weakening.

Otherwise $W \cap Z(A)$ and $W \cap Z(B)$ are proper affine subspaces covering W . Each has at most half its points. Both must therefore have exactly half, and they must be disjoint. They are parallel affine hyperplanes of W : two hyperplanes with different directions would intersect. Choose an affine u that is zero on the first and one on the second, and extend it from W to the ambient affine space. Then

$$A \models D \vee (u = 1), \quad B \models D \vee (u = 0).$$

The two weakenings and a resolution give D . This argument uses that the full Boolean assignment space is $\mathbb{F}_2^v$. $\square$

For size accounting, replace every clause's falsifying affine system by a basis, using at most $v + 1$ equations, or by the single inconsistent equation in the tautology case. Semantic equivalence preserves the soundness of all inferences. Apply Lemma 6.4 afterward. Thus S proof nodes can be normalized with at most $3S$ nodes and polynomial clause descriptions in v . Unrelated propositional variables may first be fixed to zero; semantic implication survives restriction, and a constant pivot gives a weakening. These normalizations justify using line count as the size measure in Theorem 1.1.

Theorem 6.5 (Complete clause-to-PC transfer). *A refutation of BPHP_n^{n+1} with S nodes yields, for every integer $h \geq 1$, a complete one-level old-affine family over $\mathcal{Q}_n$ with at most $3S + \binom{m}{2}$ blocks and an ordinary-PC refutation through*

$$D = \max\{2h + \ell, 4h + 1\}. \tag{6.5}$$

After normalization, the total input and coefficient inventory is polynomial in S, n, h . There is no proof-height restriction.

Proof. Introduce the compact initial linear clauses

$$C_{ii'} = \bigvee_{t=1}^{\ell} (b_{it} + b_{i't} = 1).$$

Each $C_{ii'}$ semantically implies each label-specific initial clause for the same pair in (1.1): both rows encoding the same label falsifies $C_{ii'}$. Prepend at most $\binom{m}{2}$ compact clauses and obtain each needed CNF clause by one weakening, using its existing initial node. Normalize the remaining proof as above. The total node count is at most $3S + \binom{m}{2}$.

Assign fresh blocks to its nonempty clauses. For a compact initial clause set $g_t = b_{it} + b_{i't}$ and $T_t = \prod_{s < t} (1 - g_s)$. Its value has the ordinary identity

$$P_{C_{ii'}} = E_{ii'} P_{C_{ii'}} + \sum_{t=1}^{\ell} T_t (g_t P_{C_{ii'}}). \quad (6.6)$$

The first term uses the old degree- ℓ collision axiom; the other terms use the block's companions. Every multiple has degree at most $2h + \ell$. This is an NS, hence PC, derivation at that degree.

Follow the normalized source proof in its acyclic order. Initial values are obtained from (6.6) and Lemma 6.3. Each remaining inference is simulated by Lemmas 6.2 and 6.3. Their premise values are actual earlier derived lines, so the same global ceiling (6.5) persists along the DAG. The final empty clause has value one. All blocks have inputs affine in the old v bits and disjoint fresh coefficients. Their full companion and coefficient-domain families are present. No balancing or multiplication of an entire preceding derivation is used. $\square$

7 Closing the proof-size parameters

Proof of Theorem 1.1. Fix $K > 0$ and suppose that an arbitrarily large $n = 2^\ell$ has a refutation with $S \leq n^K$ nodes. Choose $h = 3\ell$. Theorem 6.5 gives a complete one-level affine family with a PC refutation through

$$D = \max\{7\ell, 12\ell + 1\} = 12\ell + 1.$$

After scalar cleanup let $M \geq 1$ bound its number of proper blocks. We may choose

$$M \leq 3S + \binom{n+1}{2} + 1 \leq 3n^K + n^2 + 1.$$

Put

$$k = \left\lceil \sqrt{(n+1) \ln(4M)} \right\rceil, \quad B = k(D+1) = k(12\ell+2).$$

Since K is fixed,

$$k = O_K(\sqrt{n \log n}), \quad B = O_K(\sqrt{n}(\log n)^{3/2}) = o(n). \quad (7.1)$$

For all sufficiently large n , uniformly over the alleged source proofs, this implies

$$D \geq 2h + 1, \quad 1 \leq k \leq B \leq n/2, \quad k \leq n + 1, \quad n \geq 2B - 1, \quad 4(k-1) < n.$$

Theorem 5.4 excludes exactly the PC refutation supplied by Theorem 6.5, a contradiction. The threshold can depend on K but not on the chosen proof. As K was arbitrary, the asserted superpolynomial lower bound follows. $\square$

7.1 What the conclusion uses

The source clauses may have any affine rank. The argument partitions the actual complete block family by rank, uses one common multiplier on the high side, and packs every block on the low side. The final contradiction is the impossibility of an old consequence $f \neq 0$ in a separated row-linear space. It is not an application of a degree lower bound for one alone.

Only the old functional system uses PC/NS equality, established by prescribed-moment extension. The augmented system is handled by a primitive PC replay. Similarly, only the number of source blocks and their compact input descriptions need be polynomial. Expanded PC polynomials and their derivations need not have polynomial monomial count: the contradiction uses degree.

The value B in (7.1) is sufficient for the claimed qualitative theorem. This draft does not assert a sharper explicit exponential size bound.

7.2 Relationship to the broader PHP project

This result arose within a broader mathematical research project led by Kamil Braun, whose main objective is a superpolynomial lower bound for ordinary PHP in every fixed-depth $\text{AC}^0[p]$ -Frege system. That objective remains open. Theorem 1.1 concerns a specific subsystem over $\mathbb{F}_2$. The project’s research record, supporting code, and autonomous-assistant workflow are publicly available in the `math-research` repository, a versioned archive hosted on GitHub [Repo]:

<https://github.com/kbr-/math-research>

Section 8 describes the archive and framework.

The standard substitution from unary PHP into bit PHP replaces a unary variable by the conjunction describing one bit label; see, for example, [BB21]. For bounded-depth Frege it has polynomial size overhead and a constant formula-depth increase. But an affine expression in unary variables becomes a polynomial in bit conjunctions, which is generally nonlinear. This direct substitution therefore does not transfer the present theorem to unary $\text{Res}(\oplus)$, and a subsystem lower bound does not by itself establish a Frege lower bound.

8 Evidence and the research framework

The supplementary research record is maintained as a dated online notebook:

<https://kbr.is-a.dev/math-research/>

It presents arguments, failed attempts, and computation records from the broader project. The GitHub repository introduced above stores its source, supporting programs and data, and earlier versions. These materials document how the result was developed; the mathematical argument is presented in this manuscript.

8.1 Mathematical provenance and component checks

This manuscript expands the [notebook theorem and linked proof](#) recorded on 14 September 2026, in the archived repository version identified by Git commit [08dcee3](#). Its accompanying [internal publication audit](#) and subsequent literature notes in the notebook distinguish the claimed unrestricted result from the regular and depth-restricted literature.

The repository retains exact primitive-PC traces for the rule simulation, including missing-premise controls. A separate matching-moment checker stores sparse separating functionals and every potentially nonzero marginal equation for finite instances, together with falsifying perturbations. These are component checks; they do not constitute a computer verification of the asymptotic theorem. The present derivation is analytic and does not depend on rerunning those finite experiments.

8.2 An inspectable workflow for autonomous research

The project also develops an open framework for autonomous mathematical research: a collection of instructions, tools, and record-keeping conventions for AI assistants, distributed in the same repository. Its purpose is to let an autonomous assistant continue a mathematical investigation across finite context windows and session boundaries while retaining inspectable evidence [Repo]. Its main mechanisms are:

- **Persistent context.** A stable restart guide, a concise living overview, and a claim index recover the relevant definitions and proofs without reloading the full conversation or depending on a particular machine.
- **Recorded successes and failures.** Dated, append-only entries retain arguments, assumptions, obstructions, and unsuccessful attempts with explicit claim status. Corrections receive new entries instead of silently replacing history.
- **Reproducible experiments.** Protected execution, complete outputs, provenance hashes, and timing records preserve what was actually computed under bounded resources.
- **Repeated investigation and process review.** Each cycle targets a named obligation, checks its outcome, and assesses wasted work or missing tools. Bounded framework improvements are recorded separately from mathematical results.
- **A live, versioned notebook.** Saved versions in Git, a version-control system, and the public notebook let readers inspect the development locally or online.

The durable record uses ordinary source, HTML, Markdown, and data files. These mechanisms make work recoverable and reviewable; they do not make a recorded proof automatically correct or formally verified.

8.3 Contributions and development history

The principal models used were Claude Sonnet 5, Claude Fable 5.1, Claude Opus 5, and GPT-6 Astra. GPT-6 Astra was the main contributor to the mathematical development and drafting of this result. The following account describes the author’s experience; assessments of model performance are his impressions from this investigation.

The author, Kamil Braun, began in a Claude conversation on the \$20 Pro plan, initially attempting to attack $P \neq NP$. He used Sonnet 5; Fable models were unavailable to him on that plan. As the conversation progressed, the objective narrowed to the less ambitious, though still ambitious, problem of superpolynomial lower bounds for ordinary PHP in fixed-depth $AC^0[p]$ -Frege. Repeatedly reaching the five-hour usage limit, and dissatisfied with Sonnet 5’s capabilities for this task, he upgraded to the Max 20x plan.

He then attempted research with Fable 5.1. Hypotheses and proposed lemma proofs accumulated, but no clear route to the goal emerged. During the conversation, Claude began classifying his

requests as unsafe cybersecurity research; the author reports that the interface repeatedly switched him to Opus 4.8. He also attempted to continue with Opus 5, but remained dissatisfied with its performance. Increasingly frustrated, he purchased OpenAI’s ChatGPT Pro x20 plan to try GPT-6 Astra.

In the browser-based ChatGPT interface, GPT-6 Astra appeared much more confident and optimistic than Opus 5 and produced dozens of proposed lemmas and theorems. Two obstacles became apparent. First, it was unclear whether this growing body of mathematics advanced the main goal or merely supplied additional results without a sufficient application. Second, context compaction caused important information, including previously derived results, to be lost from the model’s working context. The author therefore sought an explicit context management strategy outside the chat interface. He asked ChatGPT to prepare a handoff package for Codex, an AI coding-agent environment, collecting the results and arguments developed in the conversation so that work could continue from files.

The author created a Git repository and specified requirements for the research framework described above. Using Codex CLI with GPT-6 Astra at Max thinking effort, he refined the framework through several rounds of interaction. Its first version was preserved in the initial commit of the public `math-research` repository, [2f9568f](#).

Initially, the author manually requested individual research turns, each followed by a request to identify friction in the framework and implement improvements. After several such turns, the need to automate this cycle became clear, particularly when he needed to step away from the laptop. The resulting *Spin* prompt, introduced in commit [5570c3e](#), instructs the agent to repeat research, record its results, assess the process, and improve the framework when warranted. The author assigned this continuing task through Codex’s `/goal` feature. At the time this manuscript version was prepared, the agent was continuing these cycles autonomously under that assignment.

The author’s main contribution is the design, direction, and iterative refinement of the framework, rather than the mathematical derivations themselves. Explicit resume protocols restore context, durable records preserve results and failed attempts, and computation tools support mathematical reasoning with software. The author continues to provide strategic questions and steering prompts when he notices weaknesses in the process. For example, he questioned whether successive lemmas were approaching the main goal or merely orbiting it. This concern is recorded in the notebook entry “[Classify column-literal selectors and realize arbitrary joint profiles within literal OR syntax.](#)” The resulting reassessment and route audit appear in commits [432269f](#) and [603a99d](#). Commit [e25983e](#) revised the workflow to require each research cycle to address a named missing implication, explain how its task could test or discharge that obligation, and state a concrete stopping point. The Git history and notebook research record preserve these framework changes and the strategic questions that motivated them.

The mathematical claims retain working-proof status pending external scrutiny. The classical topological input is attributed to [\[BLVZ94\]](#); the product-extension methodology is attributed to [\[BIKPRS\]](#). Original manuscript material is distributed under the [Creative Commons Attribution 4.0 International license \(CC BY 4.0\)](#).

A The required chessboard homology

We include the homological input used in Theorem [3.2](#). All homology in this appendix has coefficients in $\mathbb{F}_2$. A finite simplicial complex is called q -acyclic if its reduced homology vanishes in every degree from -1 through q . Thus (-1) -acyclic means nonempty. Below degree -1 the condition is vacuous.

Nonempty here means having at least one vertex.

Lemma A.1 (Homological cover lemma). *Let a finite simplicial complex K be covered by subcomplexes $L_1, \dots, L_t$, and let $\mathcal{N}$ be the nerve of this cover. Fix $q \geq 0$. If every nonempty intersection of s members is $(q - s + 1)$ -acyclic and $\mathcal{N}$ is q -acyclic, then K is q -acyclic.*

Proof. Here is the chain-complex comparison behind the assertion. Form the finite double complex

$$D_{a,b} = \bigoplus_{\substack{J \subseteq [t] \\ |J|=a+1, L_J \neq \emptyset}} C_b(L_J; \mathbb{F}_2), \quad L_J = \bigcap_{j \in J} L_j, \quad a, b \geq 0.$$

The vertical differential is the simplicial boundary and the horizontal differential omits one cover index and includes the chain in the larger intersection. In characteristic two their sum is a differential. Augment the horizontal rows to $C_b(K; \mathbb{F}_2)$. For each individual simplex of K , the set of cover members containing it is a nonempty simplex of indices. Its augmented chain complex is exact. Taking the direct sum over simplices proves that every augmented horizontal row is exact. Thus computing the total complex first in the horizontal direction gives the homology of K .

Compute instead first vertically. The first page consists of $\bigoplus_{|J|=a+1} H_b(L_J; \mathbb{F}_2)$. For $b > 0$ and $a + b \leq q$, these groups vanish by the assumed $(q - a)$ -acyclicity. For $b = 0$ and $a \leq q$, each nonempty L_J is connected, so the horizontal chain groups are precisely those of the nerve. In degree $a = q + 1$, the augmentation $H_0(L_J) \rightarrow \mathbb{F}_2$ is still onto for every nonempty intersection. Hence the incoming boundaries contain all the nerve boundaries needed in degree q . It follows that the second page has zero reduced groups in total degrees at most q , since the nerve is q -acyclic. No later page can create a group where the preceding one is zero. The double complex is finite, so its filtration converges and the same vanishing holds for K . Augmenting degree zero throughout gives the assertion for reduced homology as well. $\square$

Theorem A.2 (Homological chessboard bound [BLVZ94, Theorem 1.1]). *For $a, b \geq 1$, let*

$$\nu(a, b) = \min \left\{ a, b, \left\lfloor \frac{a + b + 1}{3} \right\rfloor \right\}.$$

Then the chessboard complex $\Delta_{a,b}$ is $(\nu(a, b) - 2)$ -acyclic. In particular,

$$b \geq 2a - 1 \implies \tilde{H}_{a-2}(\Delta_{a,b}; \mathbb{F}_2) = 0.$$

Proof. The stated homological bound follows from the stronger connectivity theorem in [BLVZ94]. For completeness, the standard star-cover argument gives its homological form directly as follows. Transpose the board so that $a \leq b$, and induct on a . If $\nu(a, b) = 1$, nonemptiness suffices; this includes $a = 1$. Otherwise put $q = \nu(a, b) - 2 \geq 0$, so $a \geq 2$.

For each column j , let L_j be the closed star of the vertex $(1, j)$. It consists of matchings containing that cell or extendable by it, and is a cone. These stars cover the complex: a matching avoiding row one occupies at most $a - 1 < b$ columns and can therefore be extended in that row. An intersection of $s \geq 2$ stars excludes row one and their s columns, so it is $\Delta_{a-1, b-s}$. Every proper subfamily of the b stars has nonempty intersection, while the intersection of all stars has no vertex. The nerve is therefore the boundary of a $(b - 1)$ -simplex, a sphere of dimension $b - 2$.

Write $\nu = \nu(a, b)$. For $a \geq 2$ and $a \leq b$, one has $\nu \leq b - 1$; hence the nerve is $(\nu - 2)$ -acyclic. For a nonempty s -fold intersection with $s \geq 2$, put $\mu = \nu(a - 1, b - s)$. Each entry of its defining minimum is at least $\nu - s + 1$:

$$a - 1 \geq \nu - s + 1, \quad b - s \geq \nu - s + 1,$$

and

$$a + b - s \geq 3\nu - s - 1 \geq 3(\nu - s + 1).$$

The last inequality uses $s \geq 2$. Thus $\mu \geq \nu - s + 1$, and induction shows that the intersection is $(\nu - s - 1) = (q - s + 1)$ -acyclic. If necessary transpose this smaller board before applying induction; its smaller side is at most $a - 1$. The cover lemma proves the claim. Finally $b \geq 2a - 1$ makes $\nu(a, b) = a$, giving the required special case. $\square$

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